

Particulate Matter Monitoring & Health Effects

Issue 2026-08-13 | Entry window 13 August 2026 | 17 records carried, 12 rejected

17RECORDS CARRIED
(12 REJECTED)**8**SUBTOPIC CLUSTERS
REPRESENTED**4**EFFECT ESTIMATES
WITH INTERVALS**9**INSTRUMENTATION /
EXPOSURE RECORDS**0**METADATA-ONLY
(NO ABSTRACT)

Signal of the day

A low-cost sensor calibration is only valid inside the box it was trained in — and the failure mode differs by which wall you cross.

Benjamin et al. (*Sensors*, Hannigan group at Colorado) ran a multi-season co-location of a Plantower PMS5003 and an Alphasense CO-B4 against Federal Equivalence Method monitors, then deliberately trained calibration models on *restricted* ranges of relative humidity, temperature and pollutant concentration and scored them on data outside those ranges. Three results, and the third is the one that is not obvious: within the training envelope, bias error was minimal; extrapolating outside it introduced **bias** error; and extrapolating specifically into *higher concentrations* than the training data increased **random** error.

That asymmetry matters operationally. Bias is in principle correctable after the fact if the excursion can be characterised; inflated variance at high concentration is not, and high concentration is exactly where a network is asked to support an exceedance call or a smoke advisory. The design rule follows: a training set must span the *upper* tail of the expected concentration distribution even more than it spans the humidity range, because the humidity penalty appears as a correctable offset and the concentration penalty appears as noise. Transferability claims for LCS calibration need stating as an envelope, not as an R^2 .

What changed today

- **An unusually instrumentation-heavy day: 9 of 17 records.** After a fortnight dominated by burden estimates and reviews, today's harvest is led by sensing and exposure work — LCS calibration envelopes, a self-regenerating acoustic particle sensor, a broadband-sensor-array approach to VOC speciation, gradient-boosted PM_{2.5} forecasting, Pandora column NO₂ sub-pixel variability, and a satellite-monitoring scoping review for Africa.
- **Component-resolved PM keeps outperforming mass — twice, independently.** Xu et al. report cerebral hypoperfusion ORs of **4.344** (3.323–5.734) for PM_{2.5} and **4.396** (3.327–5.869) for black carbon in 967 patients with symptomatic intracranial stenosis, with organic matter and BC together carrying **86% of the WQS mixture effect**. Tan et al. find the PM_{2.5} component mixture associated with lower eGFR in 8,477 Wuhan subway workers (WQS OR **1.183**, **1.106–1.266**). Both used the TAP component product; neither could measure components directly. The measurement gap this exposes is now the recurring theme of this watch.
- **Sensor fouling gets an engineering answer.** Fawaz et al. actuated a water droplet with Rayleigh surface acoustic waves to sweep collected particles off the sensing zone of SAW microparticle sensors inside a cascade impactor, for both NaCl solution and SiC particles, without degrading sensor performance. Surface saturation is the limiting failure mode for gravimetric-equivalent sensors in long deployments; a self-regenerating prototype is a materially different maintenance model.
- **Senolytics reverse PM-induced endothelial activation in vivo.** Gomes et al. exposed C57BL/6J mice to concentrated ambient particles and observed increased leukocyte rolling and adhesion on intact vasculature plus senescence of peripheral mononuclear and endothelial progenitor cells. Dasatinib + quercetin reversed the senescence markers (β -galactosidase, SASP genes) and restored EPC tube-forming capacity. Mechanistic rather than translational — but it places senescence upstream of early atherogenesis rather than downstream of it.
- **Africa's satellite-monitoring evidence base is 26 records.** Gould & Jagannath's PRISMA-ScR scoping review found 26 eligible studies linking satellite-derived air quality to health outcomes across the entire continent, mostly post-2015, mostly ecological, concentrated in West Africa. First record in this watch to put a number on a gap its geography figure has shown as an empty bar since 27 July.
- **No metadata-only records today** — first clean issue in five. The 08-08 to 08-12 Elsevier backlog is unchanged and stays queued for Saturday.

Corpus analytics

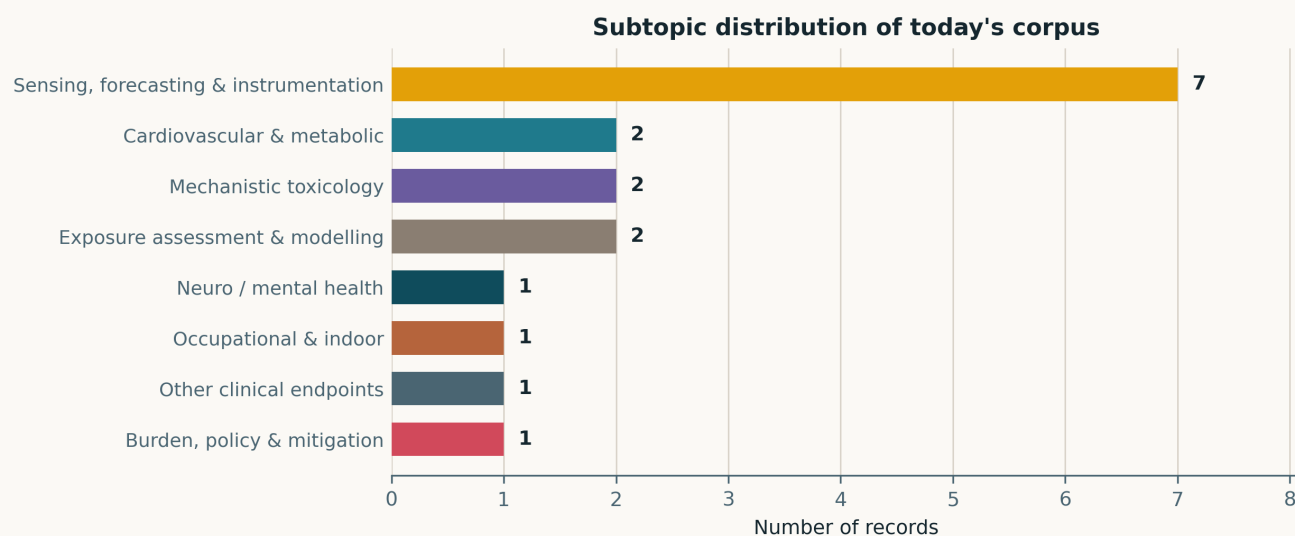


Fig. Subtopic assignment of the 17 carried records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* is the largest cluster at 7 records — the highest single-day count this watch has recorded — and with *Exposure assessment* accounts for 9 of 17. *Respiratory & allergic* and *Reproductive & developmental* are both empty today.

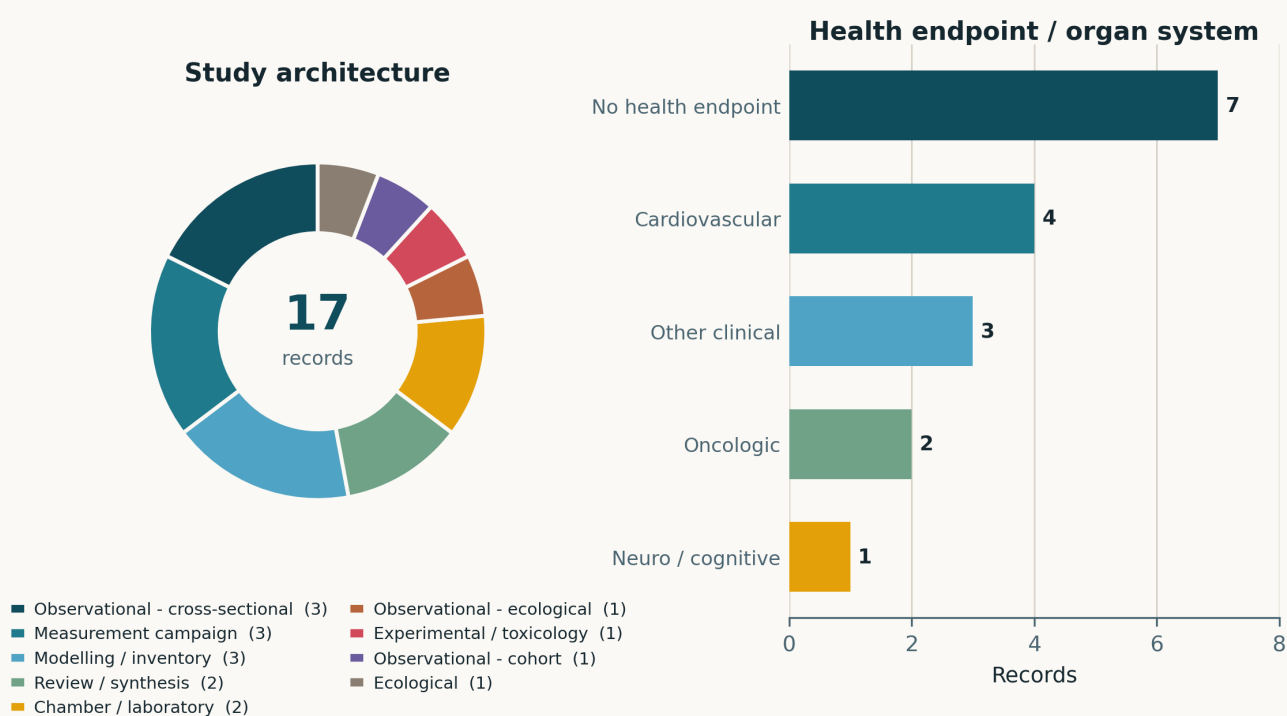


Fig. Left — study architecture. Modelling and algorithm work plus instrument development together dominate; no randomised or interventional design appears. Right — health endpoint. Nine records carry no human health endpoint, which is the direct consequence of the instrumentation share.

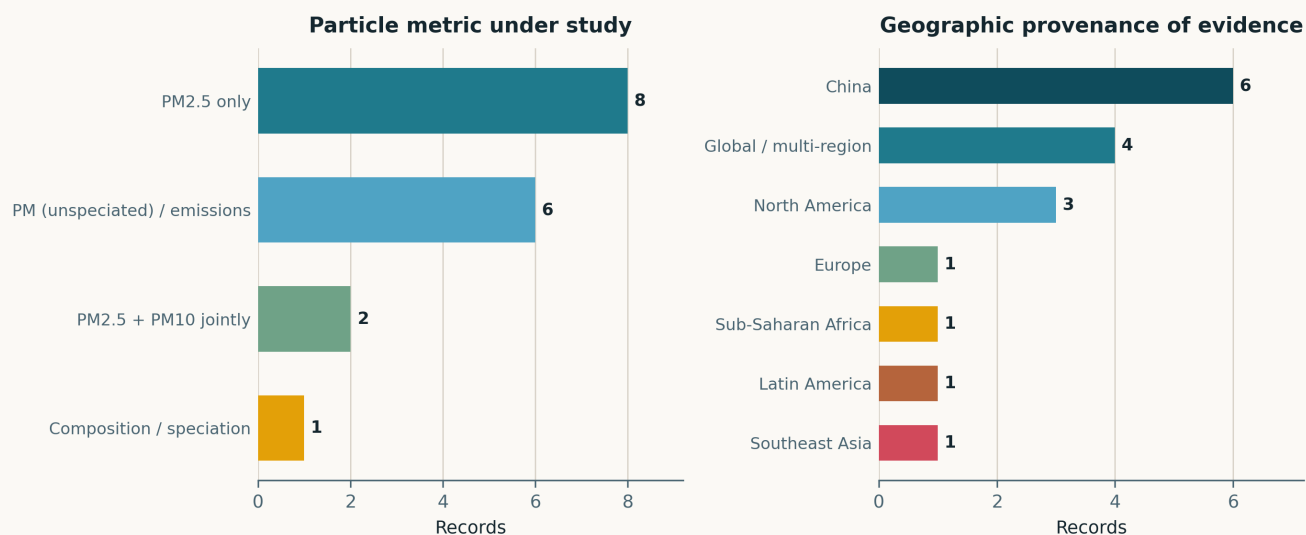


Fig. Left — particle metric. Note how many of today's exposure metrics are not PM mass: total column NO₂, aerosol depolarization ratio, VOC sensor response, a smoke plume in an image. On an instrumentation-heavy day the corpus stops being about mass concentration almost entirely. Right — geographic provenance. China contributes six records; a Research Square scoping review is the only African-focused record and it is a review of how little there is.

Life-course windows addressed today (records may span several stages)

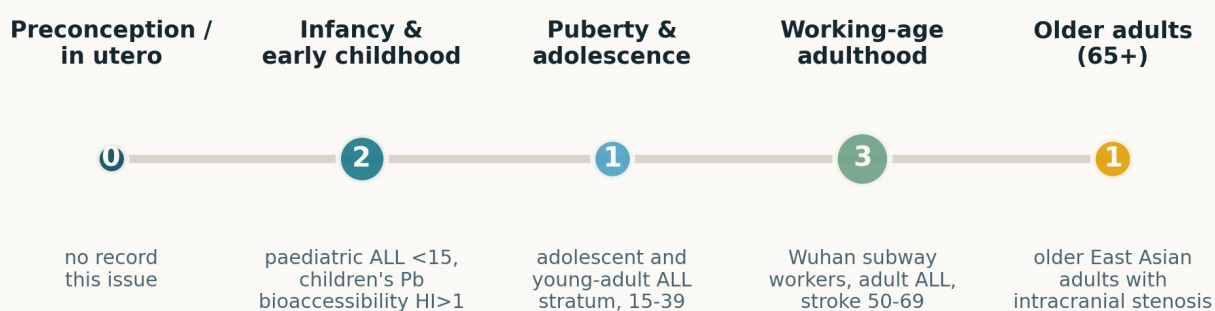


Fig. Life-course windows addressed. Counts are non-exclusive and describe this issue only. The Thai leukaemia cohort is the only record spanning three stages, and it is the reason the adolescent window is non-empty at all. The in-utero window is empty for the second consecutive issue.

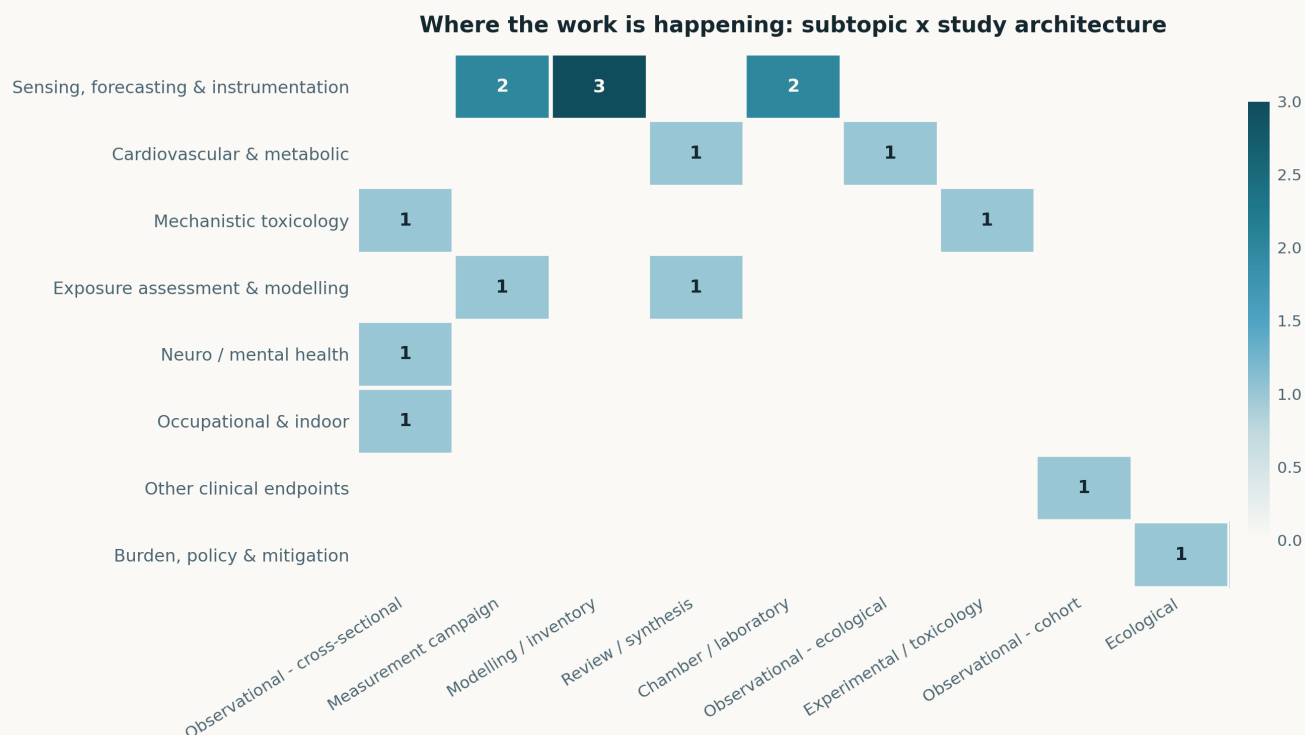


Fig. Subtopic × study-architecture cross-tabulation. The silo the 12 August issue flagged is if anything sharper today: every *Sensing* record sits in a methods, instrument or modelling column and none carries a health endpoint, while every health record sits in an observational column and none reports its own measurement.

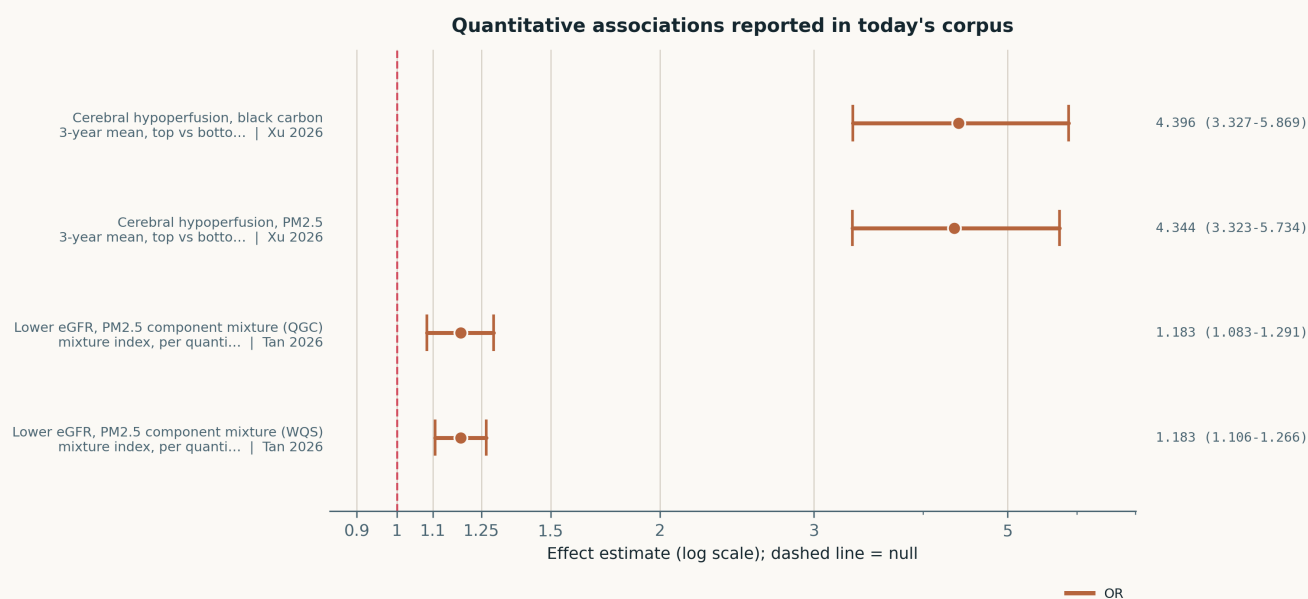


Fig. The four interval estimates carried today. **The two pairs are on completely different exposure contrasts and must not be compared to each other:** Xu et al.'s ORs are top-versus-bottom contrasts on a 3-year mean in a clinical population with established intracranial stenosis — a selected, high-risk sample in which effect sizes above 4 are not implausible but are not population estimates — while Tan et al.'s are per-quantile mixture-index ORs from two different mixture estimators (WQS and QGC) fitted to the same data, so they are not independent of each other either. Xu et al.'s component-specific ORs for sulfate, organic matter, nitrate and ammonium are reported in the abstract without intervals and are therefore not plotted.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

7 records

Benjamin et al. 2026 Sensors (Basel)

Limitations of environmental extrapolation in low-cost carbon monoxide and PM_{2.5} sensors

Multi-season co-location of a Plantower PMS5003 and an Alphasense CO-B4 against Federal Equivalence Method monitors, with calibration models deliberately trained on restricted ranges of relative humidity, temperature and pollutant concentration and then scored outside those ranges. **Within-envelope bias was minimal; crossing any envelope wall introduced bias; crossing the upper concentration wall specifically increased random error.** The design is the contribution — most co-location papers report a single in-sample R^2 and say nothing about the envelope. Limitations: one site, one sensor model per pollutant, and no aerosol-composition contrast, so the concentration effect cannot be separated from a possible composition shift at high concentration.

doi: [10.3390/s26154988](https://doi.org/10.3390/s26154988)

Fawaz et al. 2026 Sensors (Basel)

Surface cleaning of SAW-based microparticle sensors in a cascade impactor via SAW-induced droplet displacement

Rayleigh waves generated on a lithium-niobate chip actuate a water droplet that sweeps deposited particles out of the sensing zone of surface-acoustic-wave microparticle sensors integrated into a cascade impactor. Optimal droplet size and required RF power were determined; the droplet displaced both NaCl-solution residue and SiC particles without degrading sensor performance. Surface saturation is the binding constraint on long unattended deployment of mass-sensitive acoustic sensors, so a self-regenerating cycle changes the maintenance model rather than the sensitivity. Laboratory demonstration only — no ambient aerosol, no duty-cycle endurance data, and no accounting for what the droplet does to semi-volatile mass.

doi: [10.3390/s26154666](https://doi.org/10.3390/s26154666)

Gao et al. 2026 Atmos Meas Tech

Feasibility of an air sensor array for real-time detection and characterisation of VOCs

Commercial VOC air sensors are broadband — each returns one scalar reflecting an ill-defined sensitivity envelope, which is why they have seen little research use. This paper tests whether an *array* spanning three fundamentally different sensor types can recover speciation from the joint response. The measurand is VOCs, not PM, and it is carried here for the method: the same argument applies directly to optical PM sensors whose single scalar folds together size distribution, refractive index and hygroscopic growth. Feasibility study — laboratory characterisation, and the practical question of whether the array's discrimination survives ambient interferent mixtures is not settled.

doi: [10.5194/amt-19-5337-2026](https://doi.org/10.5194/amt-19-5337-2026)

Wu & Qu 2026 Environ Monit Assess

Interpretable short-term PM_{2.5} forecasting using meteorology–pollution coupling across Beijing stations

420,768 hourly observations from 12 Beijing stations, 2013–2017, under a strict chronological train–validation–test split with Diebold–Mariano testing, feature-group ablation and station-holdout transfer. At the 6 h horizon CatBoost gave **RMSE 51.24 $\mu\text{g}/\text{m}^3$, MAE 30.87, $R^2 = 0.617$** , a 6.73% RMSE reduction over persistence; station-holdout LightGBM reached 53.11 $\mu\text{g}/\text{m}^3$, 5.65% below persistence. Dominant predictors were recent PM history, wind speed and direction, diurnal cyclicity and the temperature–dew-point gap. **The honest part is the framing:** a 6.7% gain over persistence is a modest operational return, performance degrades during high-pollution episodes, and the authors state that the 2013–2017 regime would require retraining for contemporary deployment.

doi: [10.1007/s10661-026-15799-9](https://doi.org/10.1007/s10661-026-15799-9)

Adams et al. 2026 Environ Sci Atmos

Sampling sub-pixel scale variability in total column NO₂ within a coastal urban environment

Long-term Pandora spectrometer records of column-integrated NO₂ at Boston University and Harvard University, used to characterise variability within a single satellite pixel. Directly relevant to any satellite–surface fusion product: sub-pixel heterogeneity is an irreducible error term in the validation of column retrievals, and two sites a few kilometres apart in the same coastal urban area are the cleanest way to bound it. Two instruments in one metropolitan area is a small basis for a general claim about pixel-scale variability, and NO₂ column heterogeneity is not a direct proxy for PM heterogeneity.

doi: [10.1039/d6ea00033a](https://doi.org/10.1039/d6ea00033a)

Liu et al. 2026c Atmos Meas Tech

Polarization calibration of spaceborne lidar using dense cirrus-scattered solar background

Solar background scattered by optically thick ice cloud provides a semi-continuous daytime polarization calibration that complements onboard pseudo-depolarizers. At 1064 nm (CATS) molecular scattering is negligible; at shorter wavelengths it polarizes the background and biases the calibration. The paper adds a vector-radiative-transfer molecular-scattering correction and shows the daytime bias at CALIOP 532 nm is $\sim 1\%$, removed by the correction to agree with onboard calibration; EarthCARE ATLID at 355 nm is treated by model calculation only. Carried at tier C: depolarization ratio drives satellite aerosol typing, which in turn feeds AOD-to-PM_{2.5} conversion, but the link to surface concentration is two inferential steps away.

doi: [10.5194/amt-19-5325-2026](https://doi.org/10.5194/amt-19-5325-2026)

Liu et al. 2026d Sensors (Basel)

LRM-YOLO: lightweight YOLOv10n model for forest fire smoke detection in UAV images

YOLOv10n with a RepViTBlock feature-extraction module, a lightweight efficient convolutional detection head, and MPDIoU bounding-box regression loss. Against baseline: **−36.7% parameters, −42.3% GFLOPs, +1.3% mAP50, +3.7% mAP50-95, recall +3.2%, precision +3.6%**, on a purpose-built UAV-perspective dataset plus Wildfire Smoke V1. Smoke detection in imagery is a plume-presence proxy, not a concentration measurement, and the self-constructed dataset means the headline gains are partly a property of the benchmark. Relevant only as an early-warning trigger upstream of a real exposure measurement.

doi: [10.3390/s26154887](https://doi.org/10.3390/s26154887)

Exposure assessment & modelling

2 records

Gould & Jagarnath 2026 Research Square (preprint)*Satellite monitoring for air quality and public health research in Africa: a scoping review*

JB1 methodology, PRISMA-ScR reporting, five databases plus grey literature to May 2026, for studies using satellite-derived air quality data against human health outcomes in Africa. **Twenty-six records met inclusion** — for an entire continent. Most post-2015, most ecological time-series or cross-sectional, concentrated in West Africa, with PM_{2.5} the dominant exposure. This is the first record in this watch to put a number on the African evidence gap that its own geography figure has shown as an empty bar every day since 27 July. Preprint, not peer reviewed; a scoping review counts literature and does not weigh it.

doi: 10.21203/rs.3.rs-10623736/v1

da Costa Cardoso et al. 2026 Environ Monit Assess*Legacy Pb–Zn mine tailings as persistent sources of bioaccessible toxic elements in urban dust*

18 urban dust samples and reference soils from Boquira, a former Pb–Zn district in northeastern Brazil; microwave-assisted digestion and XRF for Pb, Zn, Cd, Cu, Mn, Fe, Al; geoaccumulation index, contamination and enrichment factors, ecological risk index; oral bioaccessibility by dilute-nitric-acid gastric simulation. **Pb bioaccessibility ~75%**, Cu 54.8%, Mn 28.1%, Zn 27.0%. Oral ingestion dominated over dermal; non-carcinogenic HI exceeded 1 for children on Pb, and carcinogenic risk from Cd exceeded limits for both children and adults. Eighteen samples in one district, a single gastric extraction protocol without an intestinal phase, and no particle-size fractionation — which matters, because resuspension and inhalation-relevant dose are size-dependent and this design cannot resolve them.

doi: 10.1007/s10661-026-15743-x

Neuro, cognition & mental health

1 record

Xu et al. 2026 Eur J Neurol*PM_{2.5} components and cerebral hypoperfusion, with mediation by inflammatory biomarkers*

967 patients with symptomatic intracranial arterial stenosis at one Suzhou hospital; hypoperfusion quantified from multimodal CT with MISTar; 3-year PM_{2.5} and component exposures from the TAP database; generalized additive models, restricted cubic splines, weighted quantile sum for the mixture, and mediation analysis. **PM_{2.5} OR 4.344 (3.323–5.734); black carbon 4.396 (3.327–5.869)**, then sulfate 4.359, organic matter 4.278, nitrate 3.552, ammonium 3.313 (no intervals given). OM (44.2%) and BC (42.2%) carry 86% of the mixture effect. Mediation by neutrophils 5.37%, leukocytes 4.30%, monocytes 2.73% — i.e. inflammation accounts for roughly a tenth of the total effect, so the dominant pathway is unexplained. Single-centre clinical sample already selected on stenosis; ORs above 4 in that population are not transportable to general-population risk. Component exposures are modelled, not measured.

doi: 10.1111/ene.70716

Occupational & indoor

1 record

Tan et al. 2026 Nutrients*PM_{2.5} components, nutritional indices and leisure-time physical activity vs lower eGFR in subway workers*

8,477 Wuhan subway workers; PM_{2.5} and components from the Tracking Air Pollution in China dataset; creatinine-based eGFR; GNRI and PNI nutritional indices; IPAQ-derived activity. Mixture effects by WQS, quantile g-computation and BKMR. **WQS OR 1.183 (1.106–1.266) and QGC OR 1.183 (1.083–1.291)** for lower eGFR; BKMR overall effect on eGFR of -0.045 ($-0.068, -0.022$) for a 5-percentile mixture increase against medians. Subgroups by nutrition and activity differed but **no significant interaction** was found — the authors say so, and the correct reading is hypothesis generation, not effect modification. Cross-sectional, so reverse causation is unaddressed; and residential exposure is a poor proxy for the occupational exposure of people who work underground, which is the population's defining feature.

doi: 10.3390/nu18152580

Mechanistic toxicology

2 records

Gomes et al. 2026 Cardiovasc Toxicol*Senolytics reverse PM_{2.5}-induced cell and vascular dysfunction*

C57BL/6J mice inhaling concentrated ambient particles versus filtered air showed endothelial activation *in vivo* — increased leukocyte rolling on and adhesion to intact vasculature — alongside senescence of peripheral blood mononuclear cells and endothelial progenitor cells. Dasatinib + quercetin reversed the senescence readouts (β -galactosidase activity, SASP gene expression), restored EPC tube-forming capacity *in vitro*, and limited *in vivo* endothelial activation. Positions senescence upstream of early atherogenesis rather than as a late consequence. Concentrated ambient particles deliver a supra-ambient dose; the senolytic rescue demonstrates sufficiency of the senescence pathway under that dose, not its necessity at ambient concentrations. No exposure–response and no female animals reported.

doi: 10.1007/s12012-026-10171-2

Zhang et al. 2026 J Thorac Dis*Pleural anthracosis severity and the structure and function of subcarinal lymph nodes*

Thoracoscopic grading of pleural anthracosis into four grades by three thoracic surgeons with ImageJ area quantification, against paired subcarinal lymph nodes stained with HE, Masson, EVG and immunohistochemistry for Ki67, CD31, CD68 and D2-40. Black particulate accumulation in node capsule and medulla rose with anthracosis grade, but every quantitative immunostaining index **rose then fell** with advancing severity — a consistent non-monotonic pattern across proliferation, vascularity, macrophage presence and lymphatic integrity. The non-monotonicity is the finding and is also the weakness: with a graded surgical sample and no exposure history, an exhaustion interpretation and a selection interpretation are equally consistent with the data.

doi: 10.21037/jtd-2026-1274

Cardiovascular & metabolic

2 records

Chen et al. 2026 Research Square (preprint)*Compound cold wave-PM_{2.5} exposure and cardiovascular emergency ambulance calls*

82,714 cardiovascular emergency ambulance call records from Shijiazhuang, 1 January 2014 to 31 December 2023, with three cold-wave categories defined on temperature percentiles crossed with four PM_{2.5} pollution levels to construct compound exposure events; distributed lag non-linear model with quasi-Poisson regression for lagged and interactive effects. The compound-exposure framing is the contribution — most temperature × PM work tests a multiplicative interaction term rather than constructing joint exposure states. Preprint, not peer reviewed; ambulance calls are a severity-filtered and access-dependent outcome; and a 3×4 exposure grid over ten years leaves the extreme cells thin.

doi: 10.21203/rs.3.rs-10547139/v1

Gaggini & Vassalle 2026 Int J Mol Sci*Gene-air pollution interaction in cardiovascular disease*

Narrative PubMed review of combined genetic-susceptibility and air-pollution effects on cardiovascular risk, concentrating on detoxification and inflammation genes. Argues the area could refine cardiovascular risk prediction beyond traditional factors. Narrative, not systematic; gene-environment interaction in this literature is dominated by underpowered candidate-gene studies, which the review acknowledges as “lights and shadows” but does not quantify. Useful only as an entry point to the literature.

doi: 10.3390/ijms27156818

Other clinical endpoints

1 record

Traisathit et al. 2026 PLoS One*Air pollution and mortality in acute lymphoblastic leukaemia across age groups, northern Thailand*

604 ALL patients diagnosed 1999–2020 in upper northern Thailand — a region with severe seasonal biomass-burning haze — stratified into paediatric (<15, n=344), adolescent and young adult (15–39, n=152) and adult (≥40, n=108). Five-year survival 64.5%, 19.6% and 6.7% respectively; adjusted HR for death 3.76 (AYA) and 8.10 (adult) versus paediatric; diagnosis before 2010 aHR 1.38. **PM_{2.5} ≥50 µg/m³ was associated with increased mortality in paediatric patients.** The age-group HRs are prognosis, not pollution. The pollution claim rests on comparing mean concentrations between those who died and those who did not, with a threshold analysis in one stratum; no exposure window, no treatment-era adjustment beyond the pre/post-2010 term, and ambient station concentrations as individual exposure. Signal, not estimate.

doi: 10.1371/journal.pone.0355856

Burden, policy & mitigation

1 record

He et al. 2026 Neuroepidemiology*Premature ischaemic stroke burden in adults aged 50–69 in China, India and the United States*

GBD 2023 DALYs, YLLs, YLDs and population attributable fractions for the 50–69 stratum in three countries at different stages of epidemiological transition, with ARIMA projection to 2053. Trajectories diverge: China declining, India flat at a high level, the US on a high plateau; the US has shifted to a YLD-dominant (disability) composition while India remains YLL-dominant (acute mortality); China shows a near 2:1 male:female ratio driven by male smoking. **India is characterised as facing dual pressure from persistent PM_{2.5} and rising metabolic risk.** The authors are explicit that widening intervals make the 2053 horizon scenario-based rather than forecast; a 34-year ARIMA extrapolation from modelled burden estimates is presentation, not evidence. Carried for the PAF decomposition only.

doi: 10.1159/med/adwag005

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Xu et al.	Association between PM _{2.5} components and cerebral hypoperfusion: a mediation role of inflammatory biomarkers	<i>Eur J Neurol</i>	Retrospective clinical series	PM _{2.5} , black carbon, sulfate, nitrate, ammonium, organic matter	China
Cardiovascular & metabolic					
Chen et al.	Compound cold wave-PM _{2.5} exposure and cardiovascular emergency ambulance calls	<i>Research Square (preprint)</i>	Ecological panel	PM _{2.5}	China
Gaggini & Vassalle	Gene-air pollution interaction in cardiovascular disease	<i>Int J Mol Sci</i>	Narrative review	PM _{2.5} , PM ₁₀ , traffic-related pollutants	Global
Mechanistic toxicology					
Gomes et al.	Senolytics reverse PM _{2.5} -induced cell and vascular dysfunction	<i>Cardiovasc Toxicol</i>	Animal + in vitro	Concentrated ambient PM _{2.5}	United States
Zhang et al.	Impact of pleural anthracosis severity on structure and function of subcarinal lymph nodes	<i>J Thorac Dis</i>	Retrospective clinical series	Deposited carbonaceous particles	China
Occupational & indoor					
Tan et al.	PM _{2.5} components, nutritional indices and physical activity vs lower eGFR in subway workers	<i>Nutrients</i>	Cross-sectional (cohort baseline)	PM _{2.5} component mixture	China
Sensing, forecasting & instrumentation					
Adams et al.	Sampling sub-pixel scale variability in total column NO ₂ within a coastal urban environment	<i>Environ Sci Atmos</i>	Field campaign / remote sensing	Total column NO ₂	United States
Benjamin et al.	Limitations of environmental extrapolation in low-cost carbon monoxide and PM _{2.5} sensors	<i>Sensors (Basel)</i>	Calibration model comparison	PM _{2.5} , CO	United States
Fawaz et al.	Surface cleaning of SAW-based microparticle sensors in a cascade impactor via SAW-induced droplet displacement	<i>Sensors (Basel)</i>	Instrument development	Size-segregated microparticles	France
Gao et al.	Feasibility of an air sensor array for real-time detection and characterisation of VOCs	<i>Atmos Meas Tech</i>	Instrument development	VOC sensor response (non-PM)	Laboratory
Liu et al.	Polarization calibration of spaceborne lidar using dense cirrus-scattered solar background	<i>Atmos Meas Tech</i>	Method / algorithm	Aerosol depolarization ratio	Global
Liu et al.	LRM-YOLO: lightweight YOLOv10n model for forest fire smoke detection in UAV images	<i>Sensors (Basel)</i>	Method / algorithm	Smoke plume (image proxy)	China
Wu & Qu	Interpretable short-term PM _{2.5} forecasting using meteorology-pollution coupling across Beijing stations	<i>Environ Monit Assess</i>	Method / algorithm	PM _{2.5}	Beijing
Exposure assessment & modelling					
Gould & Jagarnath	Satellite monitoring for air quality and public health research in Africa: a scoping review	<i>Research Square (preprint)</i>	Scoping review	Satellite-derived PM _{2.5} and AOD	West Africa
da Costa Cardoso et al.	Legacy Pb-Zn mine tailings as persistent sources of bioaccessible toxic elements in urban dust	<i>Environ Monit Assess</i>	Field measurement	Resuspended urban dust, Pb, Zn, Cd, Cu, Mn	Brazil
Burden, policy & mitigation					
He et al.	Premature ischaemic stroke burden in adults aged 50-69 in China, India and the United States	<i>Neuroepidemiology</i>	GBD secondary analysis	Ambient PM _{2.5} (population attributable fraction)	Global
Other clinical endpoints					
Traisathit et al.	Air pollution and mortality in acute lymphoblastic leukaemia across age groups, northern Thailand	<i>PLoS One</i>	Retrospective cohort	PM _{2.5} , PM ₁₀ , O ₃ , CO	Thailand

Provenance, screening and known gaps

Window. 2026-08-13 -> 2026-08-13, contiguous with the 12 August issue; no gap. PubMed is windowed on [Date - Entry], the Crossref-by-ISSN leg on Crossref created, and the Europe PMC leg on CREATION_DATE. Three different clocks, not treated as interchangeable.

Per-leg yield. PubMed connector, two queries (health axis 10, instrumentation axis 4, one overlapping): 13 distinct, **9 carried**. Local harvester: pubmed_health 12 and pubmed_sensing 4 surfaced **4 further carried records** the connector queries missed — the Sensors papers on LCS extrapolation, SAW sensor cleaning and UAV smoke detection — which is the clearest case yet for keeping both legs. Europe PMC 2 (**2 carried**, both Research Square preprints). Crossref by ISSN 8 (**3 carried**; three of the eight were RSC front matter — contents list, front cover, back cover). arXiv returned 145 kB with nothing in scope on or after 6 August. **OpenAlex HTTP 429, 11th consecutive.** Consensus was not re-queried this issue: the 12 August sweep on the calibration axis returned only out-of-window results and this is the sixth consecutive issue with zero Consensus carry. Scholar Gateway's corpus ends May 2026 and cannot serve an August window. ClinicalTrials.gov: no PM or air-pollution registration updated in the window; trials.json unchanged.

Rejections (12). Off-topic: broiler-diet life cycle assessment; bronchoscopic lavage case series (its "particulate matter" is aspirated gastric content); tourism and health satisfaction in older Chinese adults (air quality is one covariate among many); indirect soil NO_x emission budget; ground-based hyperspectral retrieval of *riverine* particulate organic carbon; zinc-pyrithione ultrafine particle synthesis in a microchannel reactor; Bayesian latent-component analysis of Volta Lake *sediment* PAHs; UK methane isotope source attribution; sodium-layer lidar dynamics; and three RSC journal front-matter items. All logged with reasons to state/rejected.jsonl.

Two records are preprints (Research Square) and are labelled as such in the register. They are carried because both fill a documented gap in this watch — African satellite monitoring and compound cold-wave × PM exposure — and neither is used to support a numeric claim in the analytics.

One DOI warning, accepted. 10.1159/ned/adwag005 (He et al.) resolves at doi.org but has no Crossref record; the Karger ned/ prefix pattern is new. The link was confirmed to resolve before shipping.

Labels. Subtopic, design, particle-metric and geography labels are drawn only from vocabularies already mapped in plots.py and mktable.py; no new leaf label was written this issue. Assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary.

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